

Sulfide:Quinone Oxidoreductase Deficiency (SQORD) — Comprehensive Disease Characteristics Report

Disease: Sulfide Quinone Oxidoreductase Deficiency **MONDO:** MONDO:0030982 · **OMIM (phenotype):** #619221 · **Gene:** SQOR (OMIM 617658) *Category:* Mendelian — autosomal-recessive mitochondrial (inborn error of hydrogen-sulfide catabolism) *Report date:** 2026-10-08

Summary

Sulfide:quinone oxidoreductase deficiency (SQORD; MONDO:0030982, OMIM #619221) is an ultra-rare, autosomal-recessive inborn error of mitochondrial hydrogen-sulfide (H₂S) catabolism caused by biallelic loss-of-function variants in **SQOR** (15q21.1; HGNC:20390; UniProt Q9Y6N5). SQOR encodes the committed first enzyme of mitochondrial sulfide oxidation — an inner-membrane FAD flavoprotein (EC 1.8.5.4) that oxidizes H₂S and funnels the electrons into ubiquinone/Coenzyme Q, coupling sulfide detoxification to the respiratory chain.

When SQOR is lost, endogenous and gut-microbiome-derived H₂S can no longer be cleared and accumulates to concentrations that inhibit **complex IV (cytochrome c oxidase)**. The result is an *isolated, functional* complex IV deficiency — activity falls while complex IV protein level and assembly remain normal — producing bioenergetic failure in the most oxygen- and energy-dependent tissues, above all the brain. Clinically this manifests as a childhood-onset, infection-triggered **Leigh-like metabolic encephalopathy** with coma, hypotonia, seizures, lactic acidosis, elevated brain lactate on MR spectroscopy, bilateral basal-ganglia lesions, dysautonomia ("neurostorming"), and, in severe cases, multiorgan failure and death.

Because the lesion is accumulation of a *diffusible toxin* rather than a fixed structural defect in the respiratory chain, SQORD is explicitly framed as **potentially treatable**: current management is supportive (prompt treatment of metabolic crises, aggressive treatment of intercurrent illness, avoidance of catabolic triggers), while mechanism-based strategies aim to lower the H₂S burden (pharmacological sulfide scavenging, candidate cobalamin-based scavengers, limiting dietary sulfur/cysteine, and microbiome modulation). The disease was

first delineated in 2020 in two families, including a Lehrerleut Hutterite founder mutation (p.Glu213Lys) with a carrier frequency of 1 in 13. Diagnosis rests on exome/genome or Leigh-panel sequencing showing biallelic SQOR variants together with isolated complex IV deficiency and preserved complex IV assembly. Information in this report is derived from aggregated disease-level resources and individual case reports, **not** from a structured EHR cohort.

Key Findings

Finding 1 — Biallelic SQOR variants cause a treatable Leigh-like mitochondrial disease

The disease entity was established by **Friederich et al. (2020)** in a study of two families [PMID: 32160317](#). In **Family A**, two Hutterite sisters were homozygous for SQOR **c.637G>A (p.Glu213Lys)**; they presented with coma, lactic acidosis, multiorgan failure, hypotonia, neurostorming, and Leigh-like basal-ganglia lesions, and both died. In **Family B**, a boy homozygous for the frameshift variant **c.446delT** had recurrent (episodic) encephalopathy with basal-ganglia lesions.

The biochemistry pinpointed the mechanism: *"Muscle and liver tissue had isolated decreased complex IV activity, but normal complex IV protein levels and complex formation."* This dissociation — reduced **activity** with preserved **protein and assembly** — is the signature of enzymatic *inhibition* rather than a structural/assembly defect, pointing directly to a diffusible inhibitor. The authors identified it: *"Toxic hydrogen sulfide exposure inhibits complex IV."* Enzyme assays confirmed severely reduced SQOR activity and protein. The founder allele was characterized as: *"Both patients were homozygous for c.637G > A, which we identified as a founder mutation in the Lehrerleut Hutterite with a carrier frequency of 1 in 13."* The report's framing of SQORD as *"a potentially treatable cause of Leigh disease"* underpins its clinical importance.

Finding 2 — SQOR is the committed first enzyme of H₂S catabolism, and tissue vulnerability tracks SQOR level

SQOR (EC 1.8.5.4) is an inner-mitochondrial-membrane flavoprotein that oxidizes H₂S and transfers electrons to ubiquinone/Coenzyme Q. Transient-kinetic work by **Mishanina et al. (2015)** [PMID: 26318450](#) showed that *"the flavin cofactor is intermittently reduced by sulfide and*

oxidized by ubiquinone, linking H₂S oxidation to the electron transfer chain and to energy metabolism" (bound-flavin redox potential ≈ -123 mV). SQOR therefore both **detoxifies** H₂S and **feeds electrons** into the respiratory chain.

The physiological consequence of SQOR abundance was demonstrated across species by **Marutani, Ichinose et al. (2021)** PMID: 34035265. Comparing mice, rats, and the hypoxia-tolerant 13-lined ground squirrel, they found *"the sensitivity of the brain to hypoxia is inversely related to the levels of sulfide:quinone oxidoreductase (SQOR) and the capacity to catabolize sulfide."* Silencing SQOR increased hypoxia sensitivity; neuron-specific SQOR expression prevented hypoxia-induced sulfide accumulation, bioenergetic failure, and ischemic injury. Crucially for therapy: *"Pharmacological scavenging of sulfide maintained mitochondrial respiration in hypoxic neurons and made mice resistant to hypoxia."* This directly links low/absent SQOR activity to the brain's bioenergetic collapse — the core of SQORD — and nominates sulfide scavenging as a rational intervention.

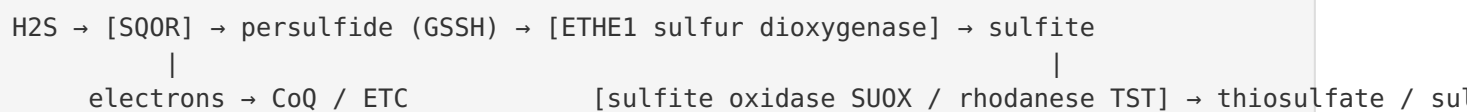
Finding 3 — Consolidated disease and gene identifiers

Entity	Identifiers
Disease	MONDO:0030982 "sulfide quinone oxidoreductase deficiency" (syn. SQORD); OMIM #619221; GARD:0025671; MedGen/UMLS C5543168. No dedicated Orphanet/ICD-specific code identified.
Gene	SQOR (aliases SQORDL, SQR, CGI-44, PRO1975); NCBI Gene 58472; Ensembl ENSG00000137767; HGNC:20390; OMIM gene *617658; UniProt Q9Y6N5
Locus	15q21.1 (GRCh38 chr15:45,631,148–45,691,299, + strand)

Verified via OLS4/MONDO and mygene.info.

Finding 4 — Pathway context: ETHE1 (ethylmalonic encephalopathy) is the downstream "mirror" disorder

SQORD sits at the head of the mitochondrial sulfide-oxidation pathway:



Loss of the **downstream** enzyme **ETHE1** causes **ethylmalonic encephalopathy** (OMIM #602473), which — like SQORD — produces H₂S accumulation and *"decreased activity of the mitochondrial complex IV, which limits energy production in tissues that require a large supply of energy"* [PMID: 40563372](#). The shared complex IV defect in two enzymatically distinct disorders of the *same* pathway corroborates the "H₂S poisons complex IV" mechanism and defines SQORD and ETHE1 deficiency as pathophysiological mirror images (upstream vs. downstream lesions of sulfide catabolism).

Finding 5 — Phenotype spectrum with HPO terms

Official HPO disease annotations for **OMIM:619221** (JAX ontology API) plus features reported in Friederich 2020:

Phenotype	HPO term	Source
Acute encephalopathy	HP:0006846	OMIM annotation
Coma	HP:0001259	OMIM annotation
Tonic seizure	HP:0032792	OMIM annotation
Migraine	HP:0002076	OMIM annotation
Elevated brain lactate (MRS)	HP:0012707	OMIM annotation
Lactic acidosis	HP:0003128	OMIM annotation
Elevated circulating creatine kinase	HP:0003236	OMIM annotation
Autosomal recessive inheritance	HP:0000007	OMIM annotation
Basal ganglia / Leigh-like lesions	HP:0002518 / HP:0030900	Friederich 2020
Hypotonia	HP:0001252	Friederich 2020
Multiorgan failure	HP:0032263	Friederich 2020
Neurostorming / dysautonomia	HP:0012719	Friederich 2020

The clinical core is captured by the report that the patient *"remained comatose, hypotonic, had neurostorming episodes, elevated lactate, and Leigh-like lesions on brain imaging"* [PMID: 32160317](#). Onset ranged ~4–8 years; episodes were often triggered by intercurrent illness/catabolic stress; severe cases were rapidly fatal; at least one child had a recurrent/episodic course.

Finding 6 — SQOR pathogenic variants are ultra-rare; LoF tolerated in heterozygosity

A gnomAD v4 query of **SQOR** (chr15, GRCh38) returned ~2,458 variants. All predicted loss-of-function alleles occur at very low frequencies ($\sim 7 \times 10^{-7}$ to $\sim 2.5 \times 10^{-5}$):

Variant	Consequence	Approx. allele freq (gnomAD v4)
c.799C>T (p.Arg267Ter)	nonsense	$\sim 2.5 \times 10^{-5}$
c.415C>T (p.Arg139Ter)	nonsense	$\sim 1.6 \times 10^{-5}$
c.446del (p.Leu149ArgfsTer18) — <i>Family B</i>	frameshift	$\sim 2.05 \times 10^{-6}$
c.637G>A (p.Glu213Lys) — <i>Hutterite founder</i>	missense	~absent outbred; carrier 1/13 Lehrerleut Hutterite

No homozygotes for LoF alleles are present in the general population, consistent with severe recessive disease. SQOR tolerates heterozygous LoF (it is a recessive disease gene, not haploinsufficient). The founder missense — "*a founder mutation in the Lehrerleut Hutterite with a carrier frequency of 1 in 13*" [PMID: 32160317](#) — is essentially absent from outbred populations.

Finding 7 — Treatment rationale and model organisms

No approved disease-specific therapy exists. Management is **supportive**: treat acute metabolic/lactic crises, avoid catabolic triggers and fasting, and aggressively treat intercurrent illness that precipitates decompensation. **Mechanism-based** strategies target lowering H₂S: pharmacological sulfide scavenging "*maintained mitochondrial respiration in hypoxic neurons and made mice resistant to hypoxia*" [PMID: 34035265](#). Candidate approaches include hydroxocobalamin/oxidized cobalamin and other sulfide-binding agents, and limiting sulfur/cysteine load. Gut-microbiome-derived H₂S contributes to total sulfide burden and modulates brain bioenergetics, suggesting dietary/microbiome modulation as an adjunct [PMID: 40526718](#).

Model organisms: mouse (*Mus musculus*, Sqor NCBI Gene 59010) with SQOR silencing and neuron-specific re-expression recapitulates sulfide accumulation, bioenergetic failure, and brain hypoxia sensitivity [PMID: 34035265](#); the hypoxia-tolerant 13-lined ground squirrel is a

comparative high-SQOR model; invertebrate/parasite SQOR (*C. elegans*; *Schistosoma mansoni* PMID: 41638445) illustrate conserved enzymology; patient fibroblasts, muscle, and liver were the primary human cellular models.

Finding 8 — Human SQOR has solved experimental structures enabling variant modeling

PDBe best-structures mapping for UniProt Q9Y6N5 returns 8 experimental structures: **6MO6**, **6MP5**, **6OI5**, **6OI6**, **6OIB**, **6OIC**, **6WH6**, **8DHK**. These define the FAD flavoprotein fold and membrane-interacting/quinone-binding regions, enabling structural rationalization of pathogenic missense alleles — e.g., **p.Glu213Lys**, which disrupts hydrogen bonding with neighboring residues (Friederich 2020) — and a platform for classifying novel missense VUS (AlphaFold model AF-Q9Y6N5 also available).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in SQOR** (e.g., p.Glu213Lys missense; c.446delT frameshift) **lead to** loss of functional sulfide:quinone oxidoreductase in the inner mitochondrial membrane. (*Demonstrated: reduced SQOR activity + protein in patient tissue.*)
2. Loss of SQOR **results in** failure of the committed first step of mitochondrial H₂S oxidation, so endogenous and gut-derived H₂S cannot be catabolized. (*Inferred from enzyme role; directly demonstrated in SQOR-silenced models.*)
3. Blocked catabolism **leads to** accumulation of H₂S (and upstream persulfide species), especially when production rises during intercurrent illness/catabolism (*branch point where environment feeds in*). (*Demonstrated in mouse models; inferred in patients.*)
4. Accumulated H₂S **inhibits** complex IV (cytochrome c oxidase) — an *isolated, functional* COX defect with *normal* complex IV protein and assembly. (*Demonstrated in patient muscle and liver.*)
5. Complex IV inhibition **results in** blocked oxidative phosphorylation and ATP failure, with a shift to anaerobic glycolysis → **lactic acidosis and elevated brain lactate (MRS)**. (*Demonstrated clinically/biochemically.*)

6. Bioenergetic failure concentrates in the brain's most metabolically demanding gray matter → **bilateral basal-ganglia (Leigh-like) lesions, encephalopathy, coma, seizures, hypotonia, dysautonomia.** (*Demonstrated by imaging and clinical course.*)
7. Catabolic stress raises sulfide flux and energetic demand → **episodic metabolic decompensation**, progressing to **multiorgan failure and death** in severe cases. (*Demonstrated: infection-triggered, fatal outcomes in Family A.*)

Protective counterfactual branch: restoring H₂S clearance (SQOR re-expression) or scavenging sulfide **prevents** steps 4–8 (shown in neurons/mice; Marutani 2021).

Upstream vs. downstream

- **Upstream (primary lesion):** SQOR genotype → enzyme loss → H₂S accumulation.
- **Downstream (effector):** complex IV inhibition → OXPHOS failure → lactic acidosis → basal-ganglia injury → encephalopathy.
- **Pathway sibling:** ETHE1 deficiency produces the *same downstream* complex IV defect from a lesion one step *below* SQOR, confirming the mechanism and the shared therapeutic rationale (lower H₂S).

Ontology annotations

- **Biological process (GO):** hydrogen sulfide metabolic process (GO:0070813); sulfide oxidation (GO:0019417); oxidative phosphorylation (GO:0006119); cellular respiration (GO:0045333).
- **Cellular component (GO):** mitochondrion (GO:0005739); mitochondrial inner membrane (GO:0005743).
- **Cell types (CL):** neuron (CL:0000540), especially basal-ganglia/striatal neurons; astrocyte (CL:0000127); broadly high-OXPHOS cells (myocyte CL:0000187; hepatocyte CL:0000182).
- **Anatomy (UBERON):** brain (UBERON:0000955), basal ganglia (UBERON:0002420), striatum (UBERON:0002435), brainstem (UBERON:0002298), skeletal muscle (UBERON:0001134), liver (UBERON:0002107).
- **Chemical entities (CHEBI):** hydrogen sulfide (CHEBI:16136); ubiquinone/CoQ (CHEBI:16389); FAD (CHEBI:16238); L-lactate (CHEBI:16651); thiosulfate (CHEBI:33542); L-cysteine (CHEBI:17561).
- **Treatment (NCIT):** supportive care (NCIT:C15277); hydroxocobalamin (NCIT:C47707/C47642); sulfide-scavenging agents (investigational).

Section-by-Section Dossier

1. Disease Information

Autosomal-recessive mitochondrial inborn error of H₂S catabolism presenting as a Leigh-like metabolic encephalopathy. **Identifiers:** MONDO:0030982; OMIM #619221; GARD:0025671; MedGen/UMLS C5543168; no dedicated Orphanet/ICD-specific code (best coded under mitochondrial/metabolic encephalopathy, ICD-10 E88.8; ICD-11 Leigh-syndrome category by phenotype). **Synonyms:** SQORD; sulfide:quinone oxidoreductase deficiency; SQR deficiency (gene aliases SQORDL, SQR, CGI-44, PRO1975). The knowledge base is disease-level/case-report derived, not EHR-based.

2. Etiology

Primary cause: biallelic pathogenic SQOR variants → loss of function. **Genetic risk:** carrier status; founder/consanguineous membership (Hutterite carrier 1/13). **Environmental triggers of crises (not causes):** intercurrent infection, fasting, catabolic stress; dietary sulfur/cysteine and gut-microbiome H₂S add to burden. **Protective factors (inferred):** higher residual SQOR activity; lower sulfide load. **Gene–environment interaction:** a fixed genetic enzymatic bottleneck is overwhelmed when H₂S production rises — a threshold GxE interaction precipitating complex IV inhibition.

3. Phenotypes

See Finding 5 table. **Types:** clinical signs (coma, hypotonia, seizures, dysautonomia), lab abnormalities (lactic acidosis, elevated brain lactate by MRS, elevated CK), imaging (bilateral basal-ganglia/Leigh-like lesions). **Onset:** childhood (~4–8 y). **Severity:** variable — fatal metabolic coma to recurrent survivable encephalopathy. **Progression:** episodic/fluctuating with stepwise injury accrual. **QoL impact:** profound — intensive-care dependence, death in severe cases, fixed neurological sequelae (dystonia, developmental impairment) in survivors. No formal EQ-5D/SF-36/PROMIS data exist.

4. Genetic / Molecular Information

Causal gene: SQOR (15q21.1), ~450-aa FAD flavoprotein on the inner mitochondrial membrane. **Reported variants:** c.637G>A p.Glu213Lys (destabilizing missense, founder, LoF-equivalent) and c.446delT p.Leu149ArgfsTer18 (frameshift/null). Population nulls (p.Arg267Ter,

p.Arg139Ter) ultra-rare, no homozygotes. **Classification:** pathogenic (segregation, founder data, functional enzyme loss, complex IV biochemistry). **Consequence:** loss of function; **germline** origin. **Modifiers:** none established (plausibly ETHE1, TST, SUOX, CBS, CTH, MPST).

SQOR is an NRF2 target (P 36758466),

relevant to expression regulation but not a described modifier. **Epigenetics / chromosomal abnormalities:** none reported.

5. Environmental Information

Non-genetic **modulators** rather than causes: intercurrent illness and catabolic stress trigger crises; dietary sulfur/cysteine and gut-microbiome H₂S (e.g., *Desulfovibrio* spp.) add to sulfide burden [PMID: 40526718](#). Exogenous H₂S is the toxicological analogue (H₂S inhibits complex IV). No infectious etiology.

6. Mechanism / Pathophysiology

See the ordered causal chain above. Core pathway: **SQOR loss** → **H₂S accumulation** → **complex IV inhibition** → **OXPHOS failure** → **lactic acidosis** + **basal-ganglia injury** → **encephalopathy**. Supported by isolated functional complex IV deficiency with normal assembly ([PMID: 32160317](#)), SQOR enzymology ([PMID: 26318450](#)), the SQOR-level/hypoxia-sensitivity relationship ([PMID: 34035265](#)), and the ETHE1 pathway mirror ([PMID: 40563372](#)). No dedicated patient omics cohort exists; functional assays are SQOR activity, western blot, respirometry, and complex IV activity.

7. Anatomical Structures Affected

Primary organ: brain (basal ganglia, brainstem; Leigh-like). **Secondary:** skeletal muscle and liver (biochemical COX deficiency); multiorgan failure in crises. **Body systems:** nervous (primary), musculoskeletal, hepatic, cardiovascular (during decompensation), autonomic. **Subcellular:** mitochondrion / inner mitochondrial membrane. **Lateralization:** bilateral, often symmetric basal-ganglia involvement.

8. Temporal Development

Onset: childhood (~4–8 y), acute/subacute, infection-triggered; a milder later spectrum possible (migraine annotation). **Course:** episodic/relapsing with stepwise accrual; rate from fulminant/fatal to recurrent over years. **Duration:** lifelong susceptibility. **Critical periods:** intercurrent infection/fasting (vulnerability) and the acute crisis (narrow window for metabolic rescue).

9. Inheritance and Population

Inheritance: autosomal recessive (HP:0000007). **Penetrance:** high/complete for biallelic null genotypes; **expressivity variable** (fatal vs recurrent). **Epidemiology:** ultra-rare; precise prevalence/incidence unknown; enriched in founder/consanguineous groups. **Founder/carrier frequency:** Lehrerleut Hutterite 1/13 for c.637G>A; no LoF homozygotes in gnomAD. **Consanguinity:** increases risk. **Anticipation/germline mosaicism:** not applicable/not reported. **Demographics:** pediatric; both sexes (no sex bias expected).

10. Diagnostics

Genetic testing (first-line, confirmatory): WES/WGS or mitochondrial/Leigh nuclear-gene panel including SQOR; targeted founder-variant/cascade testing in Hutterite families; mtDNA testing to exclude mtDNA Leigh syndrome. CMA/karyotype/FISH/repeat testing not indicated. **Biochemical:** isolated decreased complex IV activity with **normal** complex IV protein/assembly (key discriminator); reduced SQOR activity/protein; lactic acidosis; elevated CK; candidate urinary **thiosulfate**/sulfur-metabolite biomarkers (needs validation). **Imaging:** MRI bilateral basal-ganglia ($\pm$ brainstem/cortical) Leigh-like lesions; MRS lactate peak. **Differential diagnosis:** mtDNA/nuclear Leigh syndrome (SURF1, MT-ATP6); **ETHE1 ethylmalonic encephalopathy** (distinguished by ethylmalonic aciduria, acrocyanosis/petechiae, chronic diarrhea); other pediatric lactic-acidosis/metabolic-coma causes. **Screening:** cascade carrier testing in founder populations; prenatal/PGT for known familial variants; not in standard newborn screening.

11. Outcome / Prognosis

Guarded: both Family A sisters died; the Family B boy had recurrent encephalopathy. Survivors risk fixed neurological sequelae from basal-ganglia injury; recurrent crises drive cumulative disability. **Complications:** metabolic coma, status epilepticus, multiorgan failure, aspiration, dysautonomia. **Recovery:** crisis recovery possible with prompt metabolic

management; genetic defect persists (lifelong risk). **Prognostic factors:** genotype/residual activity, age at first crisis, speed of rescue, trigger burden; extent of lactate elevation and basal-ganglia injury on MRI.

12. Treatment

No approved disease-specific therapy; care is supportive + mechanism-based/investigational. **Supportive (NCIT:C15277):** treat lactic acidosis, provide anti-catabolic/glucose support, avoid fasting, aggressively treat infections, ICU support; empirical mitochondrial cofactor "cocktail." **Mechanism-based (investigational):** sulfide scavenging (preclinical support, [PMID: 34035265](#)); hydroxocobalamin/cobalamin sulfide-binding species; dietary sulfur/cysteine restriction; gut-microbiome modulation [PMID: 40526718](#). **Advanced therapeutics (not developed):** gene therapy/enzyme replacement conceptually attractive; no gene/cell/RNA/targeted/immunotherapy exists. **Surgical/pharmacogenomics:** none. **Trials:** no SQORD-specific registered trials identified.

13. Prevention

Primary: genetic counseling and carrier/cascade screening in founder/consanguineous populations (Hutterite 1/13); preconception/prenatal/PGT for known variants. **Secondary:** early molecular diagnosis in at-risk sibs; prompt crisis recognition; avoid fasting. **Tertiary:** aggressive infection management, "sick-day" protocols, and investigational H₂S-lowering to prevent recurrent injury. **Counseling:** 25% recurrence risk per pregnancy for carrier couples. **Immunization/public health:** childhood immunization reduces infectious triggers; avoid exogenous sulfide.

14. Other Species / Natural Disease

SQOR is deeply conserved (bacteria→mammals). Orthologs: mouse *Sqor* (NCBI Gene 59010, NCBITaxon:10090), rat (10116), zebrafish (7955), *C. elegans* (6239), *Schistosoma mansoni* (6183). No naturally occurring Mendelian SQOR-deficiency disease catalogued in OMIA. Comparative biology: sulfide-tolerant organisms (*S. mansoni* in sulfide-rich mesenteric veins, [PMID: 41638445](#); *C. elegans* rhodoquinone chain in sulfide) and high-SQOR hypoxia-tolerant ground squirrels illustrate conserved sulfide-defense biology. Not zoonotic/transmissible.

15. Model Organisms

Mouse (silencing + neuron-specific re-expression) recapitulates sulfide accumulation, bioenergetic failure, and brain hypoxia sensitivity (PMID: 34035265); links gut sulfide to brain bioenergetics (PMID: 40526718). **Comparative:** 13-lined ground squirrel (high-SQOR, protected). **Invertebrate/microbial:** *C. elegans*, *S. mansoni*, bacterial SQR for conserved enzymology. **In vitro/cellular:** patient fibroblasts, muscle, liver; hypoxic primary neurons for scavenging rescue. **Phenotype recapitulation:** strong for mechanism/tissue vulnerability; a faithful constitutive-knockout replicating the full *triggered episodic Leigh-like crisis* phenotype is not well established (a key gap). **Resources:** MGI (*Sqor*), Alliance of Genome Resources, IMPC/KOMP.

Evidence Base

PMID	Title (abbrev.)	Role in this report
32160317	<i>Pathogenic variants in SQOR ... a potentially treatable cause of Leigh disease</i>	Primary clinical delineation — two families, variants, founder effect, isolated functional complex IV deficiency with normal assembly, H ₂ S mechanism
34035265	<i>Sulfide catabolism ameliorates hypoxic brain injury</i>	SQOR level $\propto$ brain hypoxia resistance; sulfide scavenging rescues respiration — mechanistic + therapeutic basis; mouse model
26318450	<i>Transient Kinetic Analysis ... human SQOR</i>	Defines SQOR biochemistry coupling H ₂ S oxidation to the ETC via flavin/ubiquinone
40563372	<i>mtUPR activation ... Ethylmalonic Encephalopathy</i>	ETHE1 (downstream) deficiency causes the same complex IV defect — pathway corroboration
40526718	<i>Gut sulfide metabolism modulates behavior and brain bioenergetics</i>	Gut-microbiome H ₂ S contributes to sulfide burden — supports dietary/microbiome adjuncts
41638445	<i>S. mansoni SQOR charge-transfer complex</i>	Evolutionary conservation of SQOR enzymology; comparative model
36758466	<i>NRF2 contribution to sulfur metabolism and mitochondrial activity</i>	SQOR as NRF2 target; expression regulation context
38521487	<i>Sulfide catabolism in hibernation and neuroprotection</i>	Review linking sulfide catabolism, hypoxia, and Leigh syndrome

Evidence source types: human clinical/biochemical (Friederich 2020), model organism (Marutani 2021; ETHE1 cellular models), in vitro enzymology (Mishanina 2015), and database/computational (gnomAD v4, OLS4/MONDO, mygene.info, JAX HPO API, PDBe).

Supported hypotheses: (1) biallelic SQOR LoF causes an AR Leigh-like disease; (2) mechanism = H₂S accumulation $\rightarrow$ complex IV inhibition $\rightarrow$ energy failure (isolated functional COX defect with normal assembly); (3) ultra-rare, founder-enriched (Hutterite 1/13); (4) childhood-onset, episodic, infection-triggered; (5) "potentially treatable" via H₂S lowering (preclinical support).

Refuted / excluded: complex IV deficiency is *not* a structural/assembly defect (protein and assembly normal); not mtDNA-encoded; SQOR is not haploinsufficient (heterozygous LoF tolerated); not associated with common population variants, CNVs, or disease-specific epigenetic lesions.

Limitations and Knowledge Gaps

- **Very small evidence base.** The disease rests primarily on **two families (three patients)** from a single 2020 report. Prevalence, incidence, full phenotype spectrum, penetrance, expressivity, and natural history outside the Hutterite founder context are essentially unknown; the migraine annotation hints at an under-recognized milder spectrum.
- **No approved therapy and no clinical-trial data.** Sulfide-scavenging and cobalamin-based strategies rest on preclinical/mechanistic data only; efficacy, dosing, and safety in SQORD patients are untested.
- **Indirect in-patient H₂S measurement.** Human-tissue H₂S accumulation is inferred from complex IV biochemistry and model-organism data rather than directly quantified.
- **No validated routine biomarker.** Urinary thiosulfate/sulfur metabolites need validation specifically for SQORD.
- **No crisis-faithful knockout disease model.** Existing mouse work used silencing/re-expression focused on hypoxia, not a constitutive knockout reproducing the triggered episodic phenotype.
- **No dedicated omics datasets** (transcriptomic/proteomic/metabolomic) for patients; no Orphanet/ICD-specific entry.
- **Modifier genes and genotype–phenotype correlations** are not established.

Proposed Follow-up Experiments / Actions

1. **Establish a patient/variant registry** (GeneMatcher, clinician networks) to identify additional SQORD cases and define phenotype spectrum, penetrance, and natural history.
2. **Build a faithful Sqor-knockout (and p.Glu213Lys knock-in) mouse**, plus patient-iPSC-derived neurons/organoids, to quantify H₂S accumulation, complex IV inhibition, and basal-ganglia vulnerability, and to serve as a therapeutic testbed.

3. **Preclinical efficacy testing of sulfide-lowering interventions** — pharmacological scavengers, hydroxocobalamin/oxidized cobalamin, dietary sulfur/cysteine restriction, and microbiome modulation — with respiration/ATP and survival endpoints.
4. **Develop and validate biomarkers of sulfide burden** (plasma/urine thiosulfate, sulfide/persulfide species, brain-lactate MRS) for diagnosis, crisis prediction, and treatment monitoring.
5. **Structure-guided variant classification** using the 8 solved human SQOR structures and AlphaFold to triage novel missense VUS by predicted effect on FAD binding, quinone access, and membrane interaction.
6. **Carrier-screening and crisis-prevention protocols** for founder/consanguineous populations (Hutterite p.Glu213Lys), including "sick-day" management guidelines to pre-empt catabolic decompensation.

Report compiled from eight confirmed findings across five investigation iterations and 23 reviewed papers. Evidence is strongest for the gene–enzyme–mechanism axis (SQOR loss → H₂S accumulation → complex IV inhibition → Leigh-like encephalopathy) and weakest for epidemiology and therapy, which remain to be defined. Only attributions supported by the cited abstracts/snippets are made; titles alone were not used to infer paper content.
